

White Sheet — Cross-Population Generalization of Cardiac AI

A Nature-caliber research plan built on a real, unfilled gap (not a stack of borrowed modules)

Prepared: 3 July 2026 · **Status:** planning blueprint, pre-project · **Scope:** everything you need to (a) explain this topic to your professor without going blank, and (b) actually execute.

0. Read this first — the honest framing

Your instinct was correct, and it is the single most important thing in this document: **the previous three attempts (DEBA, CERD/SCER, the Arena one) all failed for the same reason — they invented a fancy architecture by gluing together modules that already exist in the literature, and then asserted they were novel.** That is exactly the failure mode top journals reject on sight.

- DEBA's "Bayesian Decoupling" = **logit adjustment** (Menon et al. 2021; and, in clinical form, the SCORE2 recalibration idea). Decades old.
- CERD/SCER's "Physio-Masking" = **CPR** (Causal Physiological Representation Learning). Its "Evidential Epistemic Calibrator" = standard **Evidential Deep Learning**, and the "Salient Epistemic Disentanglement" is lifted from **FetalMind (SED)**. Its "Counterfactual Agentic Decoder" = **CARE-ECG**.
- All three assumed the *answer* ("the failure is X, so we fix it with Y") before ever running an experiment.

Good science does the opposite. It finds a **real gap**, poses a **falsifiable question**, and builds the plan to *answer* it — where even a "no" is publishable. This white sheet does that.

What I can and cannot promise. I cannot promise a Nature paper — no one honestly can; Nature papers require genuine discoveries, careful execution, and some luck. What I *can* give you is: (1) a correct, verified map of the field; (2) a genuinely unfilled gap that I could not find already solved anywhere; (3) a central hypothesis that is novel, surprising, falsifiable, and globally relevant; and (4) an honest confidence rating on each Nature criterion. Where I'm uncertain, I say so.

Every paper, dataset, and code link below has been checked against the live literature (July 2026). Where the earlier documents cited things that are wrong or unverifiable, I flag it.

1. Clearing up the exact confusions your professor exposed

You went blank on three questions. Here are airtight answers. Memorize this section — it is what "knowing the topic thoroughly" looks like.

1.1 "Western models — are they LLMs? You're applying an LLM to an LLM?"

This is the question that tripped you, and the honest reason is that the *whole field uses three different kinds of model that people sloppily call "AI."* You must separate them:

Object	What it is	Examples	Does it "see" the raw ECG?
ECG foundation model (signal encoder)	A neural net trained on millions of raw ECG waveforms. Turns a 12-lead ECG into a dense vector ("embedding"). This is NOT an LLM. It's a signal/vision-style transformer or CNN.	ECGFounder, CSFM, ECG-FM, DeepECG-SSL, HuBERT-ECG	Yes — directly on the voltages.
Medical LLM (text reasoner)	A large <i>language</i> model trained on text (papers, notes). Reasons about clinical knowledge in words. Cannot read raw voltages.	Med-PaLM, Med42, MEDITRON, GatorTron	No — only text.
ECG-LLM / multimodal system	An encoder (row 1) bolted to an LLM (row 2) via a "projection layer," so the LLM can talk about a signal it never directly sees.	AMIE (cardiology), CARE-ECG, ECG-R1, KED	Indirectly — the LLM sees the <i>encoder's summary</i> , not the voltages.

So the correct framing is: **you are NOT "running an LLM on an LLM."** You are doing one of two things, and you must state which:

- **(Option A)** Auditing a *signal encoder* (an ECG foundation model) — a pure perception system. No language involved.
- **(Option B)** Auditing a *multimodal ECG-LLM system* — where a language model reasons over an encoder's output.

The plan in this document is anchored on **Option A** (the signal encoders), because that is where the demographic bias physically originates, it is mechanistically tractable, and it avoids the "LLM-on-LLM" muddle entirely. (We use an LLM only as a downstream *analysis tool* to categorize error types — never as the object under study. That distinction is crucial and it dissolves your professor's objection.)

One-line answer to give your professor: "*The 'Western models' are ECG foundation models — signal encoders like ECGFounder and CSFM, not language models. We audit the encoder directly. An LLM only appears, if at all, as a downstream reasoning layer or as an analysis tool to classify error modes; we are never stacking one language model on another.*"

1.2 "Country-specific research is low-impact for Nature"

He is right, and this is the deepest lesson. "We tested a US model on Indian data and it dropped 12%" is an **audit**, not a discovery. It is India-specific, it teaches the world nothing general about machine learning or physiology, and it has been done (in spirit) many times.

The fix is not to abandon Indian data — it's to make India **one instance of a global phenomenon**, where the *contribution is population-agnostic*. Every deliverable in this plan is a **general principle, method, or discovery** that we happen to *validate first* on an Indian cohort but that applies to any population pair on Earth. India is the test tube, not the thesis.

1.3 What actually clears the Nature Communications bar

Your professor listed the real criteria: (a) new scientific principle, (b) new framework, (c) breakthrough discovery, (d) method that changes the field's direction. The critical insight most students miss: **these are not four independent boxes you tick with four modules**. A single genuine discovery, done rigorously, tends to satisfy several at once — because a real discovery *implies* a new principle and *changes* how people work. What does **not** clear the bar is a novel-looking architecture whose parts are all borrowed. Editors and reviewers pattern-match on that instantly.

So the target is: **one real, surprising, falsifiable discovery**, cleanly demonstrated, with a mechanism and a general method that follows from it.

2. The verified state of the art (with links, datasets, code, access status)

This is your "know everything" section. All verified July 2026.

2.1 The "Western models" — ECG foundation models (signal encoders)

Model	Venue / year	Scale & data	Code / weights
ECGFounder	NEJM AI 2(7), Jul 2025.	10,771,552	Code:
	DOI 10.1056/AIoa2401033.	ECGs /	github.com/PKUDigitalHealth/ECGFounder
	arXiv 2410.04133	1,818,247 subjects, Harvard-Emory DB (HEEDB), 150 labels, RegNet encoder	Weights: huggingface.co/PKUDigitalHealth/ECGFounder

Model	Venue / year	Scale & data	Code / weights
CSFM (Cardiac Sensing FM)	<i>Nature Machine Intelligence</i> 8(2):220-233, Feb 2026. DOI 10.1038/s42256-026- 01180-5. arXiv 2507.01045	~1.7M individuals; ECG + PPG + text; trained on MIMIC-III- WDB, MIMIC- IV-ECG, CODE (Brazil) ; generative masked pretraining	github.com/guxiao0822/Cardiac
ECG-FM	<i>JAMIA Open</i> 8(5):ooaf122, 2025. arXiv 2408.05178	1.5M ECGs; Contrastive Multi-Segment Coding (CMSC) + masked reconstruction	github.com/bowang-lab/ECG-FM
DeepECG- SSL / DeepECG- SL	<i>European Heart Journal</i> 2026. DOI 10.1093/eurheartj/ehaf1119 (Avram et al.)	>1M ECGs; validated across 11 geographically diverse cohorts (881,403 ECGs)	Check paper's data-availability
HuBERT- ECG	medRxiv 2024.11.14.24317328	ECGs across 4 countries	Check preprint

Model	Venue / year	Scale & data	Code / weights

What every one of these shares (and why it matters): they are monolithic encoders trained largely on high-income-country data, and *they all demonstrably encode demographic information from the signal*. None of them separates "diagnostic morphology" from "population identity" inside the network. That is the crack you will pry open.

2.2 The ECG-LLM / reasoning space (this is CROWDED — avoid it as your core)

- **AMIE for complex cardiology** — *Nature Medicine* 32(2):616–623, Feb 2026, DOI 10.1038/s41591-025-04190-9 (O'Sullivan, Tu, et al.). Built on **Gemini 2.0 Flash**. RCT: 9 general cardiologists, 107 US genetic-cardiomyopathy cases, 3 blinded subspecialists, **10-domain rubric**. Subspecialists preferred AMIE-assisted assessments 46.7% vs 32.7% (P=0.02); fewer clinically significant errors (13.1% vs 24.3%). **System is not public**. *This is the paper your images call "the most prominent" — and its own limitations section flags that it was single-center, US, English-only*. That gap is real and citable — but the reasoning-system space around it is now saturated (below).
- **CARE-ECG** — arXiv 2604.10420, Apr 2026 (Khatibi, Wang, et al., UC Irvine/ASU). Causal graph + structural causal models + counterfactual "what-if" + causal-RAG + agentic pipeline. **This already does the "counterfactual agentic decoder" your CERD claimed as novel**.
- Plus, in 2025–2026 alone: **ECG-R1, KED** (Cell Reports Medicine), **Q-HEART, ECG world models** (ECG-WM), multiple ECG-reasoning benchmarks. Translation: **causal + counterfactual + agentic ECG-LLM reasoning is a red ocean**. Do not put your novelty there.

2.3 Datasets & benchmarks (note the Global South absence)

Dataset	Origin	Size	Modality	Link
MIMIC-IV-ECG	USA	~800k ECGs	12-lead, 500 Hz digital + EHR	physionet.org/content/mimic-iv-ecg/ (credentialed)
PTB-XL	Germany	21,799 ECGs	12-lead, SCP-ECG labels	physionet.org/content/ptb-xl/ (open)
Chapman-Shaoxing/Ningbo	China	45k+	12-lead, arrhythmia-rich	PhysioNet
CPSC-2018 / 2021	China	~6.9k+	12-lead	PhysioNet / challenge sites

Dataset	Origin	Size	Modality	Link
CODE-15% / CODE	Brazil	345k+ / millions	12-lead	physionet.org (CODE-15 open)
Georgia	USA	~10k	12-lead (PhysioNet 2021)	PhysioNet
KERALA-AF (DL study)	India	631 patients	scanned paper ECG + tabular	<i>Lancet Reg Health SE Asia</i> 2026, DOI 10.1016/j.lansea.2025.100715
BenchECG (benchmark)	Multi	—	Foundation- model eval incl. cross- population	arXiv 2509.10151

Two facts to internalize. (1) The KERALA-AF DL study is real, and it's the strongest existing Indian-ECG-DL result (AUC 0.816 for 1-yr stroke vs CHA₂DS₂-VASc 0.666; scanned paper ECGs contributed **57.1%** of predictive signal). **But it is by an Exeter/Liverpool group with Kerala clinical collaborators — it is *not* the "GMCH-TIET collaboration"** your source documents implied. Do not conflate them. (2) **Cross-country ECG generalization is already benchmarked (BenchECG) — but only among high-income countries (US, Germany, China, Brazil). India, South Asia, and sub-Saharan Africa are absent from every foundation-model benchmark.** That absence is a genuine, citable gap.

2.4 The bias / fairness literature (this is more mature than your documents admit — read carefully)

This is the part that kills the naïve versions of your idea, so you must know it cold:

- **Models encode demographics from the signal.** Age and sex are recovered with high accuracy from ECG (and even from radar cardiac signals). Race is recovered *weakly and partly via confounding* (in echo, tuning a confounder like sex swings "race AUC" from 0.57→0.84 — i.e., much apparent race-detection is *not* real race signal). Refs: multiple; echo study arXiv 2207.06421.
- **Race-ECG differences are largely NON-genetic/environmental.** *npj Cardiovascular Health* 2025 (DOI 10.1038/s44325-025-00087-1), ~10M ECGs: a CNN distinguishes Black vs White healthy subjects at AUC ≈0.86, and the signatures **appear after birth** → environmental, not innate. **This directly undermines the "stable benign ethnic phenotype" story your documents lean on.**
- **The "demographic shortcut" is already a known paradigm — in imaging.** Ghassemi/MIT (chest X-ray): demographic-prediction accuracy *correlates with* the fairness gap → models use demographics as a diagnostic shortcut; and their **training-time debiasing (subgroup robustness, group-adversarial) did not robustly transfer.**

Also: "shortcut testing" (*Nature Communications* 2023, DOI 10.1038/s41467-023-39902-7); Gichoya race-detection (*Lancet Digital Health* 2022).

- **Demographic disentanglement/suppression for ECG is already done.** VAE + dual-discriminator to suppress demographics while keeping diagnosis (medRxiv 2026.01.28.26345049). → your CERD's "**disentangle demographics from diagnosis**" is not novel.
- **Fair-ECG architectures already exist.** DA-GAT-v2 (*Scientific Reports* 2026, DOI 10.1038/s41598-026-54206-8): FiLM + graph attention for sex/age fairness, with ablation-style "mechanistic" claims. (Adjacent to you, but training-time, within-dataset, sex/age only — *not* your angle.)

The blunt implication: "disentangle demographics," "recalibrate priors," "add uncertainty," "counterfactual reasoning" — **all done**. If your plan lands in any of these, you will be scooped or rejected. The next section is where the actual opening is.

2.5 The methodological toolkit that already exists (these are your *tools*, not your contribution)

- **Causal attribution of performance drops.** arXiv 2512.09094 (Dec 2025): causal graph + Shapley values to attribute a medical-imaging performance drop to **acquisition vs annotation** — but only for imaging segmentation, only those two mechanisms, and it explicitly does **not** separate benign-population variation from true pathology shift.
- **Dataset-shift identification.** arXiv 2411.07940: separates prevalence vs covariate shift — but **explicitly states it cannot distinguish population from acquisition shift** (a named open limitation).
- **Concept-shift feature attribution.** SGShift, arXiv 2505.20634 (tabular).
- **Sparse Autoencoders (SAEs) for mechanistic interpretability.** Applied to LLMs, protein LMs (ESM-2), genomics (Evo 2), chemistry, **medical imaging** (arXiv 2603.23794), and **EEG foundation models** (arXiv 2605.13930, May 2026 — found *age-pathology entanglement* and did concept steering). **Not yet applied to ECG foundation models tied to cross-population failure.**
- **Surgical, inference-time concept removal.** LEACE (Belrose et al. 2023, closed-form concept erasure), activation steering, representation editing. Mature — **but only in LLMs and diffusion models. Never applied to medical time-series encoders for cross-population debiasing.**

Key takeaway: the *tools* to do something genuinely new all exist and are proven elsewhere. Nobody has *combined the causal question, the mechanistic dissection, and the surgical fix* on an ECG foundation model for cross-population transfer. That combination — pointed at a specific, falsifiable hypothesis — is the opening.

3. The real, unfilled gaps (ranked, with evidence)

After an exhaustive check, here are the gaps I could **not** find already solved. These are the

raw material of a Nature-caliber contribution.

- **Gap A — Nobody has causally *decomposed* the cross-population ECG gap.** The field says "models fail on non-Western data" and lumps everything together. But that gap is a *sum* of at least five distinct causes: (1) **prevalence/base-rate shift**, (2) **acquisition/device shift**, (3) **diagnostically-irrelevant benign morphology shift**, (4) **true pathology-manifestation shift** (disease genuinely looks different), and (5) **label/ontology shift** (different countries define "abnormal" differently). *No one has partitioned a real ECG gap into these.* The closest work (arXiv 2512.09094) is imaging-only and handles just acquisition vs annotation.
 - **Gap B — Nobody knows the *mechanism*.** All bias evidence is *behavioral*: train a demographic classifier, correlate its accuracy with the fairness gap, draw a saliency map. **No one has opened an ECG foundation model to *localize the shortcut* to specific internal features/directions and *causally prove* (via intervention) that they drive the errors.** The EEG SAE paper shows this is now possible; it just hasn't been done for ECG or for cross-population transfer.
 - **Gap C — There is no robust *fix*.** Every mitigation in the literature is **training-time** (adversarial suppression, reweighting, FiLM, fine-tuning), and the imaging results show these **do not transfer robustly** and trade off accuracy. **No one has demonstrated a surgical, label-free, inference-time correction** that fixes cross-population transfer without retraining and without touching the diagnostic backbone.
 - **Gap D — The field's core premise is *untested*.** Everyone *assumes* cross-population failure = "the model misperceives a benign ethnic variant as disease." **But an equally plausible mechanism has never been tested against it:** the model correctly perceives the morphology, then applies the *wrong learned prevalence prior* because it silently detects "this is a [population] ECG." These two mechanisms imply *opposite* fixes, and **nobody has designed the experiment to distinguish them.** Given the npj finding that the differences are non-genetic/environmental, the second mechanism is arguably *more* likely — which would overturn the field's working assumption.
 - **Gap E — Global South data absence in foundation-model benchmarks.** A resource gap: no Indian/South Asian (or African) cohort appears in any ECG foundation-model benchmark, digital *or* paper-based.
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4. The proposed program — the actual white sheet

Here is one coherent flagship program. It is built around a **single falsifiable hypothesis**, it delivers a **discovery + mechanism + general method**, it is **population-agnostic** (India is instance #1, not the thesis), and — critically — **it produces a strong paper even if the central hypothesis turns out false.** I deliberately do *not* wrap it in a grand acronym; the science is the point. (If you want a working name, use something descriptive later — e.g., "Causal-Mechanistic Debiasing of Physiological Foundation Models." Resist the urge to brand a module stack.)

The Prevalence-Proxy Shortcut Hypothesis. *The cross-population failure of ECG foundation models is not primarily a failure to perceive benign morphological variants. It is a localizable, surgically-correctable representational shortcut: the model has learned an internal pathway from **population-identity features** to **disease-prevalence priors**, and when deployed on a new population this pathway fires with the wrong prior — producing systematic false positives. Consequently, the failure can be removed by a targeted, label-free intervention that severs this pathway, **without** retraining and **without** collecting more diverse data.*

Why this is a *real* scientific claim (and not another module stack):

- **Falsifiable.** It makes a sharp prediction that can come out false: if severing the internal population→prevalence path leaves the errors intact, the hypothesis is wrong and the failure really *is* perceptual. Either outcome is a publishable finding.
- **Surprising / consequential.** If true, it overturns two entrenched assumptions at once: (i) that you need ever-larger, perfectly diverse global datasets to fix bias, and (ii) that the failure is benign-variant misperception. It replaces the "data arms race" with "find and cut the shortcut."
- **Global.** The mechanism, if it exists, is population-agnostic — the *same* circuit-type fails on *any* under-represented population.

4.1 Stage 1 — DECOMPOSE: what is the gap actually made of?

Question: Of the total cross-population performance gap, how much is prevalence shift vs acquisition vs benign morphology vs true pathology-manifestation vs labeling?

Why it's novel: No one has partitioned an ECG cross-population gap into these five causes. This alone (done rigorously) is a contribution, and it *sets up* the hypothesis test.

Method (concrete):

1. Write down an explicit **causal graph** of ECG generation: disease (Y) → morphology; population/environment (E) → (benign morphology, prevalence ($P(Y|E)$), and confounders); acquisition (A) → signal rendering; annotation practice → observed ($label$).
2. Build **matched sub-cohorts** to hold specific factors constant (e.g., same disease + same acquisition, vary population; or same population, vary device).
3. Use **controlled interventions on the signal**: re-render an ECG in a *different acquisition style* (device transfer, paper-photo degradation) while provably preserving the physiological content — isolating ($\Delta_{acquisition}$). Digital↔paper is directly relevant to global deployment (and connects to the KERALA-AF finding).
4. Attribute the gap with a **Shapley-style decomposition over mechanisms** (extending arXiv 2512.09094 from "acquisition vs annotation" to the full five-way clinical

taxonomy — the key extension being the **benign-vs-pathological morphology axis**, which the imaging work cannot do).

Predicted discovery (Gap D): benign-variant *misperception* is a *small slice*; **prevalence-proxy behavior + labeling + acquisition dominate**. If confirmed, that is your "breakthrough discovery."

Deliverable: the first quantitative partition of an ECG cross-population gap — a result the whole field currently lacks.

4.2 Stage 2 — LOCALIZE: where is the shortcut inside the model?

Question: Is there an identifiable internal pathway (population-features → prevalence-prior), and does it *causally* drive the false positives?

Why it's novel (Gap B): first mechanistic (circuit-level) dissection of an ECG foundation model tied to cross-population failure. All prior ECG bias work is behavioral.

Method (concrete):

1. Train **Sparse Autoencoders (SAEs)** on the residual stream / activations of an *open-weight* encoder (ECG-FM is ideal; ECGFounder weights are open too) — the same recipe proven on EEG (arXiv 2605.13930) and medical imaging (arXiv 2603.23794).
2. Use **linear probes** to find directions/features that encode *population identity* and, separately, that shift *predicted prevalence*.
3. Run **activation patching / causal mediation**: patch the population-features from a target-population ECG into a source-population ECG (and vice versa) and measure the change in the false-positive rate. This is the causal proof that the population→prevalence path *drives* the errors (not just correlates).
4. Decode the identified directions back to **physiologically interpretable signatures** (which leads, which segments), as the EEG SAE paper did with spectral decoding — so a cardiologist can see *what* the shortcut is keying on.

Deliverable: a causally-validated internal circuit/subspace — the mechanism the field has never had.

4.3 Stage 3 — SEVER: fix it without retraining, without labels

Question: Can a surgical, inference-time edit that removes the prevalence-proxy direction restore cross-population performance — beating training-time debiasing — without harming diagnostic accuracy?

Why it's novel (Gap C): surgical concept-erasure/steering has never been applied to a medical time-series encoder for cross-population debiasing, and no *label-free, inference-time* fix exists that beats training-time methods.

Method (concrete):

1. Apply a **closed-form concept-erasure edit (LEACE-style)** or a **steering vector** that removes/neutralizes the identified prevalence-proxy direction — **at inference, weights frozen**.
2. Add a **physiological-preservation constraint**: the edit must *not* erase genuinely diagnostic morphology (verified against Stage 2's decoded signatures). This is the technically interesting part — surgically removing the *spurious* prevalence pathway while provably preserving the *causal* morphology pathway.
3. Make it **label-free**: the erasure direction is estimated from *unlabeled* target-population ECGs only (you never need target-population diagnostic labels — this is what makes it deployable in low-resource settings and what answers your professor's "how do you make it generalize" question).
4. **Head-to-head benchmark** against: fine-tuning, group-adversarial suppression, reweighting, FiLM. Metric that matters: **the equity gap** (e.g., ΔFPR on benign-variant-carrying ECGs), *not* just aggregate AUROC — because a model can have fine average AUROC while systematically over-calling disease in one population.

Deliverable: a general, deployable, label-free correction — a "field-changing method" candidate *if it wins*.

4.4 Stage 4 — GENERALIZE: prove it's a law, not a one-off

Question: Does the same circuit-type and the same fix generalize across populations *and* across models?

Why it matters: this is what converts "an interesting result on one model/population" into "a general principle" — the difference between a solid paper and a Nature paper.

Method:

- Replicate across ≥ 2 **population pairs**: (i) an Indian cohort (instance #1 — see data section), and (ii) a *fully public* pair you can run *today* without any private data, e.g., **CODE-15 (Brazil) $\leftrightarrow$ MIMIC-IV-ECG (US) $\leftrightarrow$ PTB-XL (Germany) $\leftrightarrow$ CPSC (China)** via BenchECG-style transfer; plus African data if obtainable.
- Replicate across ≥ 3 **foundation models**: ECG-FM, ECGFounder, CSFM.
- If the same population $\rightarrow$ prevalence shortcut appears and the same surgical fix works across all of them $\rightarrow$ you have a **general principle of physiological foundation models**, not a curiosity.

Deliverable: the generality evidence that underwrites the "new principle" and "changes the field" claims.

4.5 Rigor & evaluation protocol (do not skip — this is what reviewers attack)

- **Pre-register** the hypothesis and the decomposition analysis *before* looking at outcomes. This is your strongest defense against "you fished for a story" — the exact critique that sinks the previous three attempts.

- **Primary metric = the equity gap** (e.g., $\Delta\text{FPR} / \Delta\text{FNR}$ across populations, and specifically false-positives on ECGs carrying benign variants like early-repolarization, high-voltage LVH-mimics, anterior T-wave inversion), reported alongside AUROC, calibration (ECE), and Equalized Odds.
 - **Clinical grounding:** use an AMIE-style blinded rubric, but *repurposed* to score the thing that matters here — "did the system over-call pathology on a benign regional variant?" — evaluated by cardiologists.
 - **Negative-result plan (write it now):** if Stage 2 finds no clean circuit, or Stage 3's surgical fix fails, the paper becomes "*We tested whether cross-population ECG failure is a prevalence-proxy shortcut or benign-variant misperception, and rigorously showed it is [X] — with implications for how the field should (and should not) invest.*" Still novel, still publishable. **This robustness to failure is the mark of real science and is exactly what your prior attempts lacked.**
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5. Explicit mapping to the four Nature criteria (with honest confidence)

Your professor wants each of (a)–(d) *strongly* matched, not force-fitted. Here is my honest assessment — including where I'd rate a match "moderate" rather than "strong," because pretending everything is a slam dunk is how you got three fake frameworks.

Criterion	What this program delivers	Honest confidence
(a) New scientific principle	<i>"Cross-population failure of physiological foundation models is a localizable, surgically-correctable representational shortcut — a learned population→prevalence pathway — not an irreducible data-diversity deficit or a perceptual failure."</i>	Strong — this is a genuine, general, previously-unstated principle, <i>if the mechanism is confirmed.</i>
(b) Completely new framework	The DECOMPOSE→LOCALIZE→SEVER→GENERALIZE program: the first causal decomposition of an ECG cross-population gap fused with the first mechanistic dissection fused with the first label-free surgical fix. The <i>combination</i> on a medical time-series encoder is new.	Moderate-to-strong — the individual tools exist elsewhere; the <i>framework</i> (question + method + fix, aimed at this problem) does not. Be honest with reviewers that the tools are borrowed but the synthesis and the target are new.
(c) Breakthrough discovery	The empirical finding that the dominant driver of the gap is a prevalence-proxy shortcut (Stage 1+2), overturning the benign-variant-misperception assumption.	Strong if the hypothesis holds; still a real discovery if it's falsified (you'd have shown the field's assumption is right <i>or</i> wrong, rigorously — either is news).
(d) Method that changes the field's direction	Replacing the "collect infinitely diverse data + fine-tune" paradigm with "localize and surgically sever the shortcut" — label-free, inference-time, generalizes across populations and models.	Aspirational until proven — this claim is <i>earned by Stage 3–4 results</i> , not asserted. If the surgical fix beats fine-tuning across models/populations, the claim is strong. If it only partially works, you soften it. Do not overclaim this one before the data is in.

Bottom line for your professor: criteria (a) and (c) are strongly addressed by the *question and discovery*; (b) is well-addressed by the *synthesis*; (d) is *conditional on results* — which is the intellectually honest way to present it, and it signals scientific maturity rather than hype.

6. The adaptation question (your professor's second point)

He asked: *if the model performs poorly, what exact changes make it generalize?* This program answers it directly and non-trivially:

1. **The decomposition tells you which lever to pull.** If the gap is mostly *prevalence* → recalibrate the prior (cheap, known). If mostly *acquisition* → harmonize/augment acquisition. If mostly *labeling* → remap the diagnostic ontology. If mostly *benign-morphology-as-shortcut* → the surgical severing fix. **Nobody currently knows which lever applies, because nobody decomposes the gap.** That diagnostic capability is itself valuable.
 2. **The surgical fix is the adaptation mechanism — and it beats fine-tuning on the two axes that matter in the Global South:** it needs **no target-population labels** and **no retraining**, so it works where you have unlabeled Indian ECGs but no annotated ground truth and no GPUs to fine-tune a foundation model.
 3. **Honesty on the residual:** for *genuinely novel pathology* (true concept shift — e.g., a regional disease that manifests in a truly new morphology the model has never seen), *no representation trick* suffices; you need targeted labeled examples. Say this plainly. Reviewers respect a paper that states the boundary of its method. This is the "give reasons if not possible" fallback your professor allowed — and stating it *strengthens* you.
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7. Risks, failure modes, and my honest read on the odds

- **The hypothesis might be false.** Mitigated by design: a rigorous answer either way is publishable (§4.5). This is the single biggest reason this plan is safer than the previous three.
- **Mechanistic interp might not yield clean features.** SAEs sometimes produce entangled/messy features (the EEG paper itself found "wrecking-ball" entanglement). Mitigation: even *partial* localization + a working *behavioral* intervention is a result; and the decomposition (Stage 1) stands alone.
- **The surgical fix might not fully beat fine-tuning.** Then (d) softens and the paper leans on (a)+(c). Still strong.
- **Data-access risk (load-bearing).** Your plan's Indian instance depends on a *real, ethics-approved* clinical ECG cohort. **The "GMCH-TIET collaboration" in your source documents is not the KERALA-AF group — verify what data you actually**

have, under what data-use agreement and IRB/ethics approval, before promising it. Crucially: you do not need private Indian data to start — Stage 1–4 run entirely on public cross-country data (Brazil/US/Germany/China). Indian data makes the equity story land harder but is not on the critical path for the core science.

- **Compliance:** do not route credentialed data (MIMIC, HEEDB) through commercial LLM APIs — that violates their data-use agreements and is an instant desk-reject (this was a real flaw in the DEBA plan). Keep the LLM strictly as an offline analysis tool on de-identified error summaries, if used at all.
 - **Scooping risk.** The mechanistic-interpretability-for-physiological-models space is moving *fast* (EEG SAE paper is May 2026). Move quickly on the open-weight models; the Indian-cohort angle is your differentiator because others don't have that data.
 - **My honest odds statement.** *Nature Medicine / Nature Communications* is a realistic aspiration for the full program if the hypothesis holds and the fix generalizes — but the modal outcome for most rigorous projects is a strong specialized venue (*npj Digital Medicine, European Heart Journal - Digital Health, Lancet Digital Health, or a top ML venue like NeurIPS/ICML for the methods*). Aim for Nature; do not stake your self-worth on it. A clean discovery in a strong journal beats a rejected "Nature-or-nothing" gamble every time.
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8. Concrete first 90 days (so you can actually start)

1. **Weeks 1–2 — Lock the foundations.** Download open-weight ECG-FM and ECGFounder weights + PTB-XL and CODE-15 (both open). Reproduce baseline AUROC. Confirm you can extract internal activations from ECG-FM (you'll need this for SAEs). Read, in full: the npj race-ECG paper, the Ghassemi demographic-shortcut work, the EEG SAE paper (2605.13930), the imaging causal-attribution paper (2512.09094), and LEACE.
 2. **Weeks 3–5 — Pilot Stage 1 on public data only.** Run the cross-country transfer (CODE↔MIMIC↔PTB-XL↔CPSC) and do a *first-cut* gap decomposition (prevalence vs acquisition vs the rest). This is a self-contained result and de-risks everything. **No Indian data needed yet.**
 3. **Weeks 6–9 — Pilot Stage 2.** Train an SAE on ECG-FM activations; probe for population and prevalence directions; run one activation-patching experiment to test whether the population→prevalence path moves the false-positive rate. This is the make-or-break test of the hypothesis — get to it fast.
 4. **In parallel — Nail down the Indian cohort.** Verify the *actual* data, its data-use agreement, and its ethics/IRB approval. If it's real and approved, it becomes your equity showcase (Stage 4). If it's not yet real, the public-data program still stands.
 5. **Before any "confirmatory" run — pre-register.** Write the hypothesis, the decomposition, and the primary metrics on OSF/AsPredicted. This one act is worth more to your credibility than any fancy architecture.
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9. Master reference table (verified July 2026)

#	Item	Link / ID	Access
1	ECGFounder (NEJM AI 2025)	DOI 10.1056/AIoa2401033 · arXiv 2410.04133 · github.com/PKUDigitalHealth/ECGFounder · huggingface.co/PKUDigitalHealth/ECGFounder	Weights open; HEEDB credentialed (bdsp.io/content/
2	CSFM (Nat Mach Intell 2026)	DOI 10.1038/s42256-026-01180-5 · arXiv 2507.01045 · github.com/guxiao0822/Cardiac-Sensing-FM	Code open
3	ECG-FM (JAMIA Open 2025)	arXiv 2408.05178 · github.com/bowang-lab/ECG-FM	Open weights + code
4	DeepECG- SSL/SL (Eur Heart J 2026)	DOI 10.1093/eurheartj/ehaf1119	Paper open-access; mc access varies
5	AMIE complex cardiology (Nat Med 2026)	DOI 10.1038/s41591-025-04190-9	Paper accessible; system public
6	CARE-ECG (2026)	arXiv 2604.10420	Preprint
7	KERALA-AF DL study (Lancet Reg Health SEA 2026)	DOI 10.1016/j.lansea.2025.100715	Open-access
8	BenchECG foundation- model benchmark	arXiv 2509.10151	Preprint
9	Race-from- ECG is non- genetic (npj Cardiovasc Health 2025)	DOI 10.1038/s44325-025-00087-1	Open-access

#	Item	Link / ID	Access
10	Demographic shortcut in imaging (Nat Commun 2023)	DOI 10.1038/s41467-023-39902-7	Open-access
11	Demographic suppression in ECG (VAE, 2026)	medRxiv 2026.01.28.26345049	Preprint
12	Causal attribution of imaging performance drops (2025)	arXiv 2512.09094 · github.com/PeterMcGor/CausalDropMedImg	Preprint + code
13	Dataset-shift identification (2024)	arXiv 2411.07940	Preprint
14	SAE mechanistic interp of EEG foundation models (2026)	arXiv 2605.13930	Preprint
15	SAE for medical imaging (2026)	arXiv 2603.23794	Preprint
16	LEACE (closed-form concept erasure)	Belrose et al. 2023 (NeurIPS)	Open
17	Fair-ECG architecture DA-GAT-v2 (Sci Rep 2026)	DOI 10.1038/s41598-026-54206-8	Open-access

#	Item	Link / ID	Access
18	Ethnic ECG differences / benign variants	<i>Am J Cardiol</i> 2016 (S0002-9149(16)31027-X); review PMC11942291; PMC6147850	Open/abstract
19	PTB-XL / MIMIC-IV-ECG / CODE-15 / CPSC	PhysioNet	PTB-XL & CODE-15 op MIMIC credentialed

A note on the source documents you were given

Both the DEBA and CERD/SCER PDFs contain **fabricated or unverifiable citations** (several arXiv IDs don't correspond to real papers; "Indian Heart Journal 2026" could not be verified; the AMIE author list and the "GMCH-TIET = KERALA-AF" conflation are wrong). The *real* anchors are the ones in the table above. Trust these, not the earlier documents' reference lists.

The one thing to remember: your edge is not a cleverer architecture. It is asking a **real, falsifiable question nobody has answered — *what actually causes cross-population failure, and can it be surgically removed?*** — and answering it rigorously on models and data you can actually access. That is what gets into top journals. Everything else in this document serves that one idea.